

William Otterson

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Education

Bachelor of Science, Mechanical Engineering, The University of Texas at Austin Expected May 2027

Work Experience

Mechanical Engineering Intern, International Biomedical - Austin, TX May 2026 - Present

- Created an in house computational optics tool that automated freeform lens synthesis, optical simulation, verification, and manufacturing file generation for phototherapy illumination systems
- Designed a freeform lens achieving 99.6% illumination uniformity versus 94.2% for a leading competitor lens, iterating printed prototypes in house in 1 day rather than through outside optics vendors
- Led physical design of a neonatal phototherapy device from concept to tested working prototype, selecting LEDs, optics, and thermal management components to meet clinical and regulatory requirements
- Benchmarked the market leading phototherapy device across irradiance, thermal, and power draw
- Validated the working prototype with a bili radiometer against the market leading device, delivering AAP intensive level irradiance of 30 uW/cm²/nm or greater across 99% of the treatment target
- Built FDM printed assembly jigs with GD&T callouts across 3 revisions, halving adhesive strip application time while holding repeatable bezel alignment in production
- Presented weekly design reviews and test results to engineering leadership across an 11 week program
- Sized the device power supply against a 43 W worst case budget, selecting a 60 W unit at 72% load

R&D Engineer Co-op, Smith+Nephew - Austin, TX June 2025 - December 2025

- Built an automated connector lifetime test rig with a custom KiCad PCB, a microcontroller, and pneumatic actuation, running 7,500 cycle endurance tests per connector to quantify wear and electrical failure modes
- Reduced electrosurgical controller component temperatures 5°C to IEC 60601 compliant levels, mapping thermals with 16 thermocouples and Python data logging to drive a cooling redesign
- Performed root cause analysis on contact pad contamination, tracing it to an undocumented vendor spring substitution with a degrading coating and driving a vendor corrective action
- Managed in house additive manufacturing across 7 FDM and SLA printers (Fortus, Bambu Lab, Asiga), tuning print parameters and materials to maximize part strength and throughput

Research Assistant, MINIMAX Lab - Austin, TX August 2024 - April 2026

- Developed piezoelectric microfluidic pump prototypes for ingestible drug delivery 1.6x more energy efficient than a commercial pump, iterating channel geometry in SolidWorks with custom test fixtures

Projects

TI-Nspire Wi-Fi + AI Module - github.com/OtterCadGH/Internet-Link February 2025 - April 2026

- Designed an ESP32 expansion board giving TI Nspire calculators web API access via a Lua interface
- Published open source firmware and build guides for 3 hardware revisions, growing a 36 star GitHub repo and a Discord community of users building the board from my guides

Automated Overdose Emergency Aid Device June 2024 - September 2024

- Prototyped an overdose response device fusing pulse, blood oxygen, and acoustic breathing sensing to detect an event and guide untrained bystanders through Narcan administration and emergency calls

Skills

CAD: SolidWorks, GD&T (ASME Y14.5), tolerance analysis, hand calculations, FEA, CFD

Embedded and Electronics: KiCad PCB design, Arduino, ESP32, sensor integration, pneumatics, relays

Programming and Data: Python, MATLAB, Lua, data acquisition and logging, test automation, API integration

Fabrication: FDM and SLA 3D printing, CNC machining, laser cutting, composite layups